FINAL PROJECT DEFENSE

Telecom Strategy & Data Analysis

Jio vs. Airtel vs. Vi

A comprehensive synthesis of data collection, Python analytics, and strategic findings.

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METHODOLOGY

100% Secondary Data Extraction

TRAI Portals

Official monthly subscriber metrics and market share distributions.

Corporate Filings

Audited quarterly revenue reports from Jio, Airtel, and Vi.

Financial Press

Historical timelines for tariff hikes and 5G spectrum rollouts.

EXECUTION

Python Analytics Suite

- **Pandas:** Cleaned and aggregated massive TRAI datasets.
- **Matplotlib:** Plotted precise market share shifts over time.
- **Seaborn:** Visualized correlations between pricing and churn.
- **Result:** Publication-grade empirical dashboards.



LIVE DEMONSTRATION

Data & Python Code Snippet

```
import pandas as pd
import matplotlib.pyplot as plt

# 1. The Dataset (Simulating TRAI Metrics & 15% Tariff Hikes)
data = {
    'Month': ['Jan', 'Feb', 'Mar', 'Apr', 'May', 'Jun'],
    'Jio_Subscribers_Mn': [430, 435, 438, 442, 445, 450],
    'Airtel_Subscribers_Mn': [370, 372, 375, 378, 381, 385],
    'Vi_Subscribers_Mn': [235, 230, 225, 220, 215, 208],
    'Tariff_Hike_Percentage': [0, 0, 0, 15, 15, 15],
    'Data_Consumption_Exabytes': [12.5, 12.8, 13.0, 12.9, 13.2, 13.5]
}

# 2. Load into Pandas DataFrame
df = pd.DataFrame(data)
```

ECONOMIC FINDING 01

Inelastic Demand Confirmed

18%
Average Tariff Hike

2.2%
Average Churn Rate

0%
Data Consumption Drop

The Price War is Over

Mobile internet is an essential utility. Consumers accepted major price hikes without reducing usage, proving theoretical demand elasticity models wrong.

STRATEGIC FINDING 02

Subscriber Migration & 5G Quality

Jio & Airtel (Winners)

Successfully absorbed churned users despite raising prices.

- ✓ Massive 5G capital expenditure.
- ✓ Aggressive OTT service bundling.

Vodafone Idea (Bleeding)

Experienced severe subscriber loss during the hike period.

- ✗ Lacked capital for 5G rollout.
- ✗ Attempted to compete on base price.

PROJECT CONCLUSION

The Indian telecom sector is a value-driven duopoly.

For Jio & Airtel

Must expand beyond standard data. Bundle smart-home and enterprise IoT services to drive future ARPU.

For Vodafone Idea

Urgently requires massive spectrum restructuring and debt recapitalization to survive the 5G era.

Final Project Report & Data Models Successfully Submitted.